

Yufeng (Alex) Xia

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PROFILE

- MSc by Research student at the Institute for Computing Systems Architecture (ICSA), University of Edinburgh, supervised by Prof. Mahesh K. Marina. My interest lies at the intersection of **wireless systems, systems for AI, and large-scale emulation**. My work spans **AI-RAN**, where live radio processing and data-driven workloads share the same infrastructure under real-time constraints, as well as **emulation and digital twin systems**, the platforms on which RAN and edge-AI algorithms are developed and validated at scale.

I build the layer between the radio and the model, making live network behaviour something models can be trained and validated against, and idle radio compute something they can run on. These systems have led to publications at top-tier venues including ACM MobiCom, and what interests me most now is carrying this line of work into next-generation radio access and edge AI infrastructure. **Available full-time from September 2026.**

EDUCATION

- **University of Edinburgh — MSc by Research, Computer Science** Edinburgh, UK
Supervised by Prof. Mahesh K. Marina. Thesis: Flux-DT: An Elastic, Preemption-Tolerant Digital Twin Runtime for AI-RAN Sep 2025 – Sep 2026 (expected)
- **University of Edinburgh — BSc Computer Science** Edinburgh, UK
*Grade: **First Class Honours*** Sep 2021 – May 2025
 - **Relevant courses:** Machine Learning Systems, Machine Learning Practical

TECHNICAL SKILLS

- **Languages** C, C++, Python, Java, Bash.
- **ML & GPU Systems** PyTorch, CUDA programming, GPU profiling and roofline analysis, distributed training at scale, resource scheduling and allocation policies.
- **Performance & Evaluation** Benchmarking and microbenchmarking, system bottleneck analysis, large-scale system evaluation.
- **Infrastructure** Linux, Docker, Kubernetes, Git, CMake, Microsoft Azure, CloudLab.

RESEARCH AND PROJECT EXPERIENCE

- **Flux-DT — Elastic, Preemption-Tolerant Digital Twin Runtime for AI-RAN** MSc by Research Thesis
University of Edinburgh — sole author Sep 2025 – Present
 - **Sharing RAN compute with a stateful workload:** Built a runtime that lets a radio digital twin run on bursty spare compute in the radio path, holding its scene state across interruption without violating live RAN priorities. Prior compute-sharing designs assume the co-located workload is disposable.
 - **Near-real-time control policy:** Designed and implemented the control loop that answers radio-driven resource pressure on the timescale the radio imposes.
 - **Multi-cell fidelity placement algorithm:** Formulated fidelity placement as a joint objective across co-channel cells and **designed a greedy heuristic** deciding which tiles are worth solving under interference
- **EmuRAN — Scalable, Cost-Effective Cloud-based RAN Emulation System** MobiCom 2026
Research Assistant; owned the end-to-end evaluation Conditionally Accepted
 - **Public cloud operation:** Deployed and operated the emulator on public cloud infrastructure, emulating over 10,000 mobile devices, two orders of magnitude larger in scale than existing software solutions.
 - **Fidelity and scalability validation:** Designed and ran the evaluation campaign, covering both fidelity against a real RAN and scaling behaviour under constrained compute.
- **Weaver — Foundation Model Training over AI-RAN Compute Infrastructure** Under Submission
Research Assistant; spare-compute controller evaluation and large-scale experiments 2024 – 2026

- **RAN-centric spare compute control:** Evaluated the controller deciding how much accelerator capacity an AI workload may harvest from shared RAN infrastructure, a near-real-time resource-allocation problem on GPU-accelerated hardware. On a multi-site testbed it harvests up to **83%** of spare compute without degrading live radio performance.
- **PHY experiments and large-scale training runs:** Ran PHY-layer experiments and GPU profiling to characterise the interference between training workloads and radio processing, and carried out the large-scale foundation-model training runs used in the evaluation.
- **CoSenseAQ — LLVM Auto-Quantization from Sensor Specifications** Under Submission
Research Assistant, ICSA; first author Apr 2024 – May 2025
 - **Sensor specifications as compiler input:** Built an LLVM auto-quantization framework that drives optimisation from sensor specifications, information a compiler cannot recover from source alone, reducing code size and execution time on embedded targets.
 - **Embedded evaluation:** Evaluated on embedded ARM devices, measuring speedup, code size reduction and lower power consumption.
- **Morphling — Emulator for Distributed Machine Learning at the Edge** EdgeSys 2026
Research Assistant; Best Paper Award Oct 2025 – Mar 2026
 - **Measurement-driven edge emulation:** Contributed to a measurement-driven emulator for distributed ML training on heterogeneous edge devices, supporting up to 1,800 emulated devices on a single GPU server with low latency and thermal error.

INDUSTRY EXPERIENCE

- **Senior Tech — Back-end Development Intern** Suzhou, China
Backend and QA work on Uniqlo’s logistics and mobile products Jun 2023 – Sep 2023
 - **Express delivery platform:** Contributed to the backend architecture upgrade of Uniqlo’s express delivery platform, migrating a live production system from JFinal to Spring Boot and rewriting its interface layer; took part in code review and release testing alongside the permanent team. Java, Spring Boot, MySQL, Redis, Kafka
 - **Android application testing:** Tested the Uniqlo Android app for performance, reliability and user experience, identifying and reporting defects ahead of release.

SELECTED PUBLICATIONS

- EmuRAN: Scalable and Cost-Effective Cloud-based RAN Emulation System** *MobiCom’26 (cond. accepted)*
Ujjwal Pawar, Andrew E. Ferguson, **Yufeng Xia**, Xenofon Foukas, Mahesh Marina, Bozidar Radunovic
- Weaver: Foundation Model Training over AI-RAN Compute Infrastructure** *under submission*
Leyang Xue, Tianxin Wang, Xin Zhe Khooi, Jiaxun Yang, Dheeraj Mahendiran, **Yufeng Xia**, Mun Choon Chan, Myungjin Lee, Mahesh Marina
- Morphling: Emulator for Distributed Machine Learning at the Edge** *EdgeSys’26 (Best Paper)*
Leyang Xue, **Yufeng Xia**, Ismaeel Bashir, Eren Mendi, Jiaxun Yang, Myungjin Lee, Mahesh Marina
- Wasp: Foundation Model Training System for the Edge** *SEC’26*
Leyang Xue, **Yufeng Xia**, Myungjin Lee, Mahesh Marina
- CoSenseAQ: Compiler Optimizations using Sensor Technical Specifications with Automatic Quantization** *under submission*
Yufeng Xia, Pei Mu, Antonio Barbalace

AWARDS, LANGUAGES AND SERVICE

- **Awards Best Paper Award**, EdgeSys 2026 (Workshop co-located with ACM MobiSys 2026), for *Morphling*. Student Travel Grant, ACM MobiSys 2026.
- **Languages** Mandarin (native), English (fluent). IELTS 7.0; GRE 336.
- **Service** Student volunteer, EuroSys 2026 (on-site conference operations).

REFERENCES

- **Prof. Mahesh K. Marina** University of Edinburgh, MSc by Research supervisor · mahesh@ed.ac.uk
- **Dr Antonio Barbalace** University of Edinburgh, BSc and RA supervisor · antonio.barbalace@ed.ac.uk